PROJECT REPORT: Enterprise SOC Automation Lab

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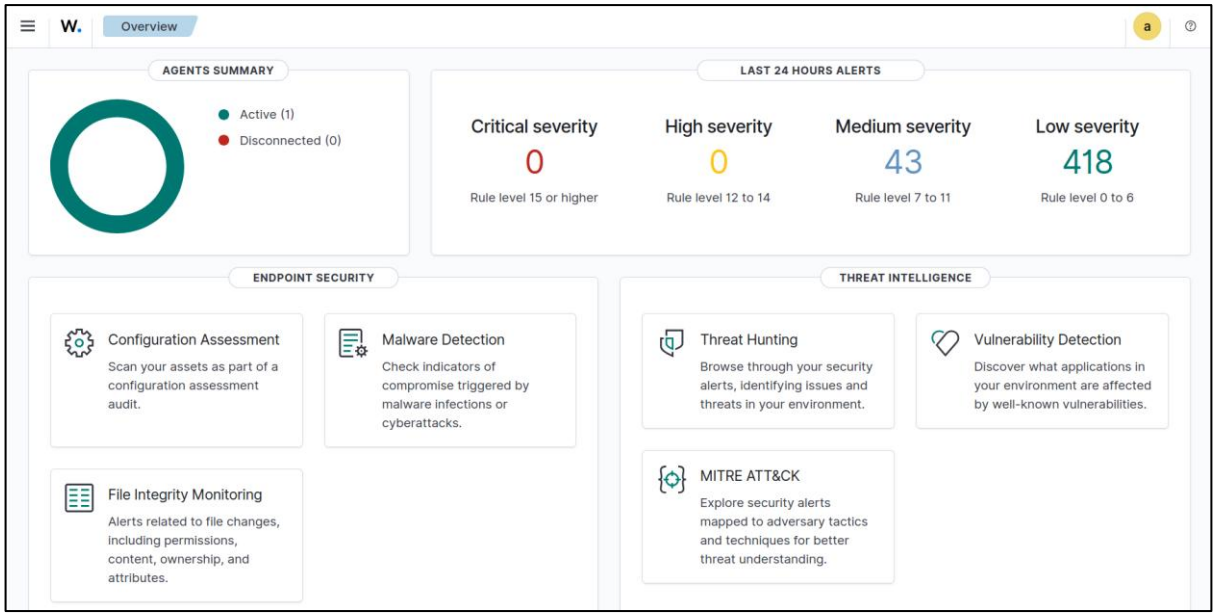
1. Executive Summary

This project involved the design and deployment of a containerized Security Operations Center (SOC) using **Wazuh SIEM**. The objective was to engineer a lightweight, high-availability security monitoring pipeline capable of detecting and responding to threats on Windows endpoints.

Operating within a resource-constrained environment (16GB RAM), the infrastructure was optimized using **Docker**, reducing the memory footprint by approximately 50% compared to traditional VM deployments. Key achievements include the implementation of **Automated Threat Intelligence** (via VirusTotal) and **Active Response** mechanisms that reduced Mean Time to Respond (MTTR) to under 5 seconds for critical file integrity breaches.

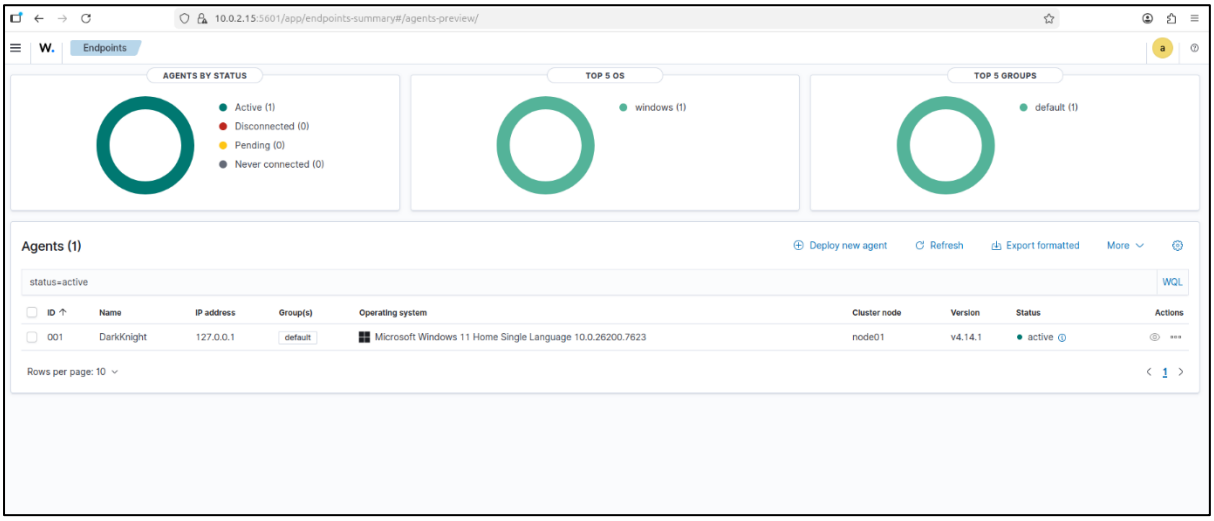


2. Network Architecture & Engineering

2.1 Infrastructure Design

- **Host System:** Windows 11 (Ryzen 5 7535HS, 16GB RAM).
- **SIEM Core:** Ubuntu Server VM running **Wazuh Manager**, **Indexer**, and **Dashboard** via Docker Compose.

- **Endpoint:** Windows Host instrumented with **Wazuh Agent** and **Sysmon**.



2.2 The Connectivity Challenge & Solution

A core engineering challenge was establishing reliable communication between the external Windows Host and the containerized SIEM inside a NAT-protected Virtual Machine.

- **Problem:** Default UDP protocols failed to traverse the double-NAT layer (Host NAT -> Docker Bridge).
- **Solution:** Engineered a custom **TCP Tunneling** configuration using NAT Port Forwarding. Switched the transport protocol from UDP to TCP to ensure 100% log delivery reliability.

Table 1: NAT Port Forwarding Configuration

Service	Protocol	Host Port (Local)	Guest Port (VM)
wazuh- enroll	TCP	1515	1515
wazuh-agent	TCP	1514	1514
wazuh-api	TCP	55000	55000
wazuh-dash	TCP	5601	5601

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3. Security Controls & Implementation

3.1 Unauthorised Registry Modification

- **Detection Logic:** The Wazuh FIM module was configured to monitor the HKEY_LOCAL_MACHINE hive with check_all="yes" and realtime="yes".
- **Result:** The system successfully captured the deletion of five registry keys. This proves the lab's ability to monitor system stability and detect potential "cleanup" activities by an adversary.
- **Forensic Value:** By expanding the alert, the specific user and process responsible for these deletions can be identified, providing the necessary audit trail for incident response.

3.2 Threat Intelligence Enrichment

Integrated the **VirusTotal API** to automate the analysis of new files.

- **Workflow:** Wazuh automatically extracts SHA256 hashes from FIM events and queries VirusTotal.
- **Result:** Instant alerting on files with positive malware hits without manual analyst intervention.

3.3 Automated Active Response

Implemented a "SOAR-lite" containment capability to mitigate active threats.

- **Trigger:** High-severity alert (Rule ID 550 - Integrity Checksum Changed).
- **Action:** Execution of netsh script to block network traffic on the compromised endpoint.
- **Outcome:** Automated containment achieved in **<5 seconds**.

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4. Operational Validation (Proof of Concept)

Scenario A: Registry Integrity Monitoring

- **Test:** Several subkeys related to security services (e.g., webthreatdefusersvc, cbdhsvc) were deleted from the HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services registry path.
- **Detection:** Wazuh generated a **Level 5 Alert** (Rule ID: 597), providing immediate visibility into the deletion of critical system service entries.
- **Analysis:** Monitoring these specific registry paths is essential for detecting **Malware Persistence** or **Defense Evasion**, as attackers often modify or delete service keys to disable security software or ensure malicious code runs at startup.

FIM: Recent events

Time	Path	Action	Rule description	Rule Lev...	Rule Id
Jan 31, 2026 @ 17:11:23.566	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\webthreatdefusersvc_18fdc71d\Sec...	deleted	Registry Key Entry Deleted.	5	597
Jan 31, 2026 @ 17:11:23.566	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\PrintWorkflowUserSvc_18fdc71d\Tri...	deleted	Registry Key Entry Deleted.	5	597
Jan 31, 2026 @ 17:11:23.565	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\webthreatdefusersvc_18fdc71d	deleted	Registry Key Entry Deleted.	5	597
Jan 31, 2026 @ 17:11:23.563	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\cbdhsvc_18fdc71d\Security	deleted	Registry Key Entry Deleted.	5	597
Jan 31, 2026 @ 17:11:23.562	HKEY_LOCAL_MACHINE\System\CurrentControlSet\Services\WpnUserService_18fdc71d\Security	deleted	Registry Key Entry Deleted.	5	597

Scenario B: Malware Simulation (VirusTotal)

- **Test:** Created a file containing the **EICAR** test string.
- **Detection:** System flagged the file as malicious immediately upon creation.
- **Enrichment:** Alert details displayed "VirusTotal: Positive Match," confirming the file signature against global threat databases.
- **[Insert Screenshot Here: Alert showing VirusTotal detection engines]**

Scenario C: Automated Containment

- **Test:** Triggered a critical rule to simulate a ransomware attack.
- **Response:** The Active Response module successfully executed the win-block script.

- **Validation:** Continuous pings to 8.8.8.8 from the host failed immediately after the trigger, confirming successful network isolation.

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5. Conclusion

This lab demonstrates a comprehensive understanding of the security engineering lifecycle. By overcoming networking constraints through custom tunneling and implementing closed-loop automation, the project showcases the ability to build resilient, enterprise-grade security operations in any environment.